Mobile Devices

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Mobile Devices - Input

Touch panel / screen

Absolute and direct device

Aligned so that it is possible to do input and output together in one place

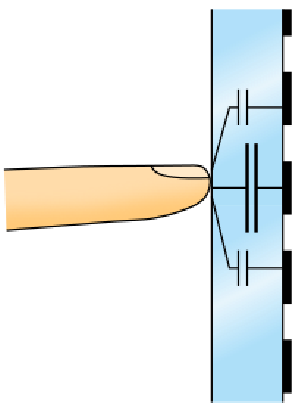
- actually point at things on the screen

Resolution limited by size of finger

- The "Fat Finger" problem
- Or requires a pen

Touch Input

Capacitive: on the electrical properties of the human body to detect when and where the user touches on a display (e.g., touch screens of mobile devices)



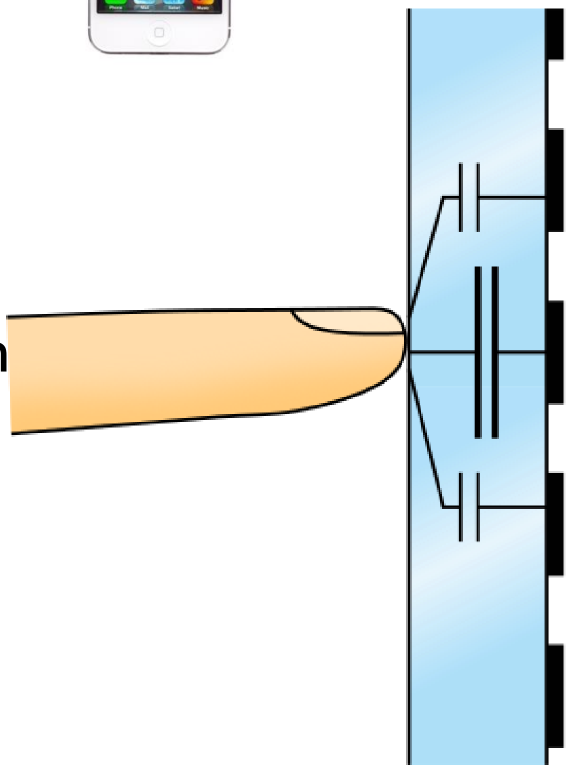
Touch screen construction

Capacitive (current best)

- iPhone et al., SmartBoards
- Multi-touch standard (your phone)

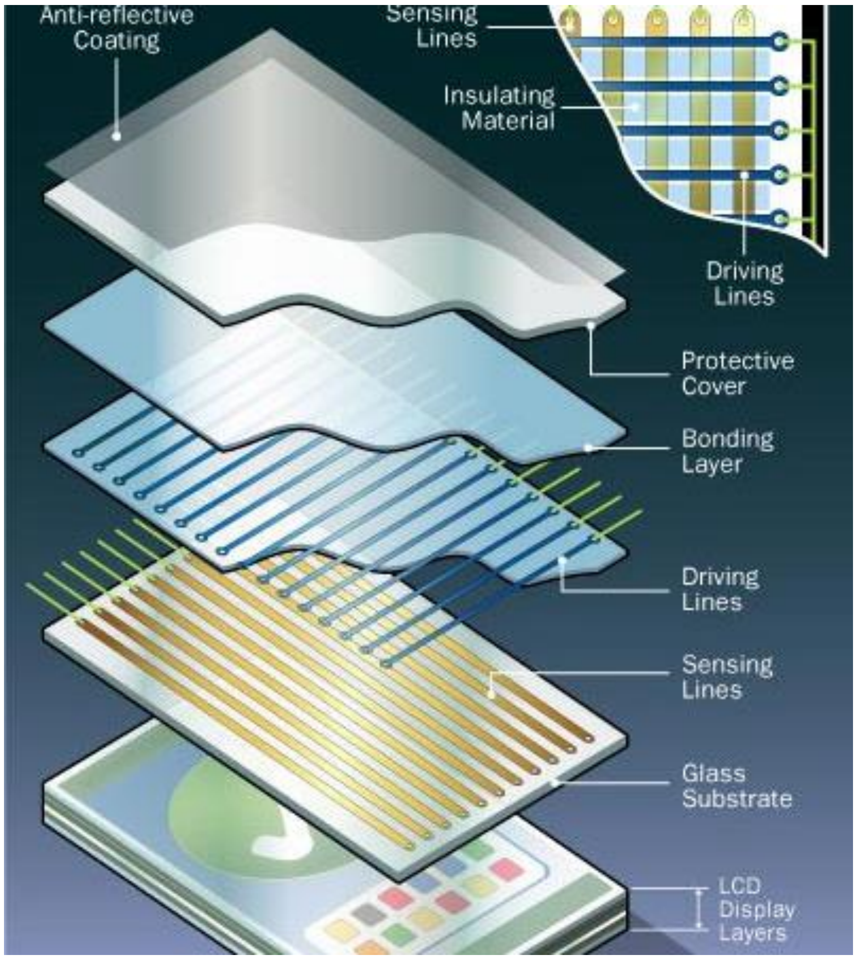
A capacitive touchscreen panel consists of an insulator such as
glass, coated with a transparent conductor such as indium tin oxide (ITO)

Touch causes a capacitor to dynamically form



https://engineering.mit.edu/ask-an-engineer/how-do-touch-sensitive-screens-work

Iphone's Capacitive Touch Screen

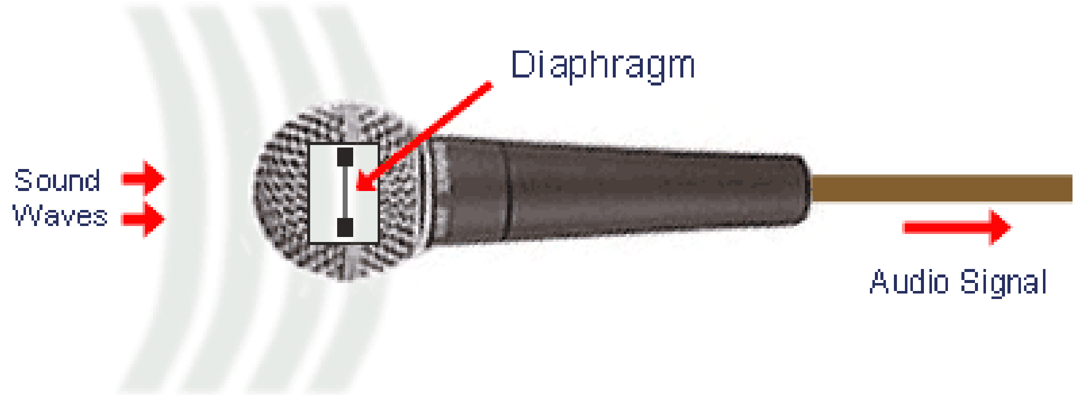


Driving lines carry current

Sensing lines sense finger

Audio Input

-Microphone:



Converts sound waves into audio signal. Diaphragm vibrations are converted into an electrical current which becomes the audio signal.

This signal is an “electrical image” representing the characteristics of the acoustic waveform. Generally, the output signal from a microphone is an analogue signal either in the form of a voltage or current which is proportional to the actual sound wave.

Audio Input

1 S**ound waves** carry energy toward the microphone.

2 **Diaphragm** moves back and forth when the sound waves hit it.

3 The **coil** also moves back and forth.

4 The **permanent magnet** produces a magnetic field that cuts through the coil. As the coil moves back and forth through the magnetic field, an electric current flows through it.

5 The **electric current** with the same frequency as sound waves flows out from the microphone sound recording device.

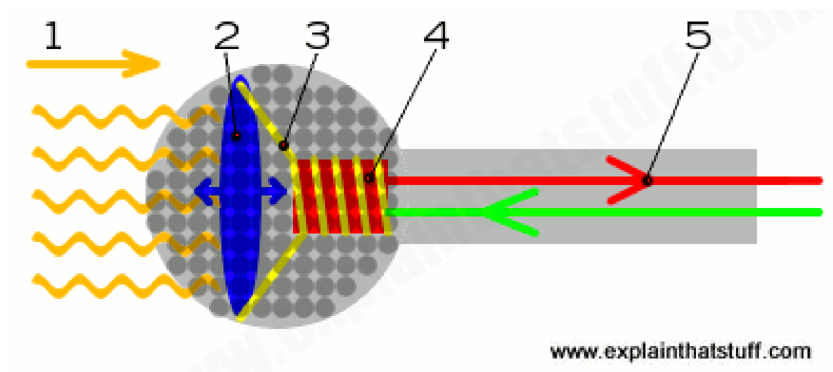
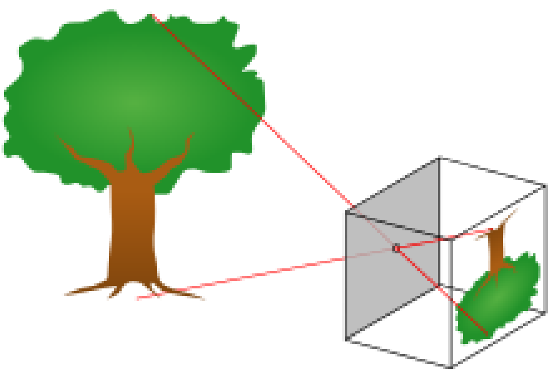


Image Input

Photo Camera: 3D to 2D



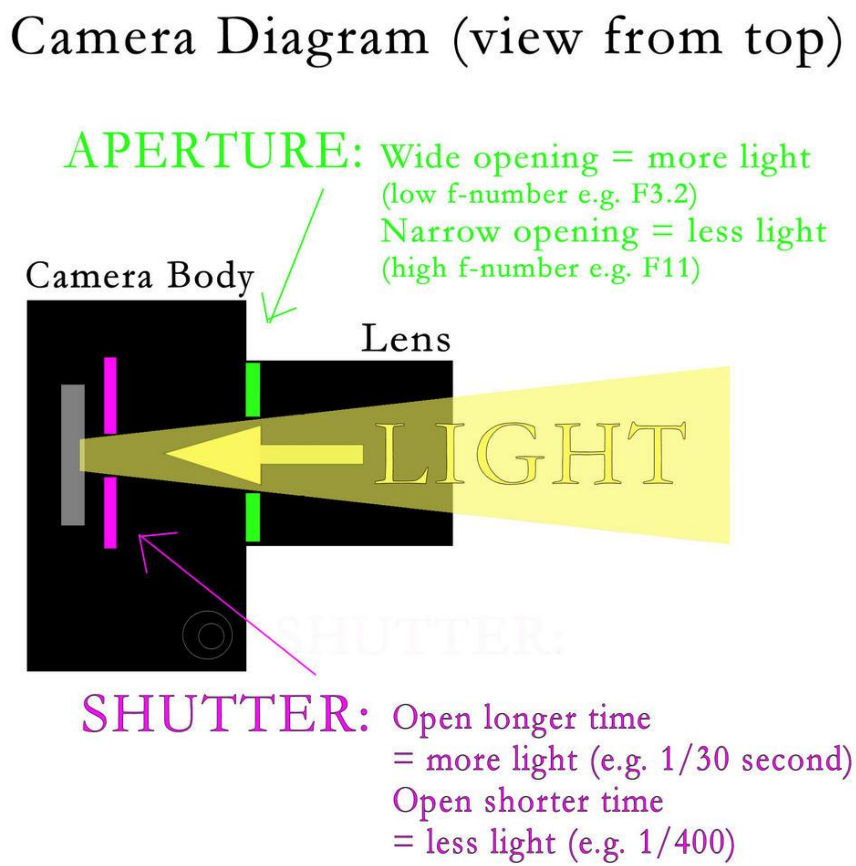


Image Input

A plastic or metal **case** that is completely light-tight to protect the film.

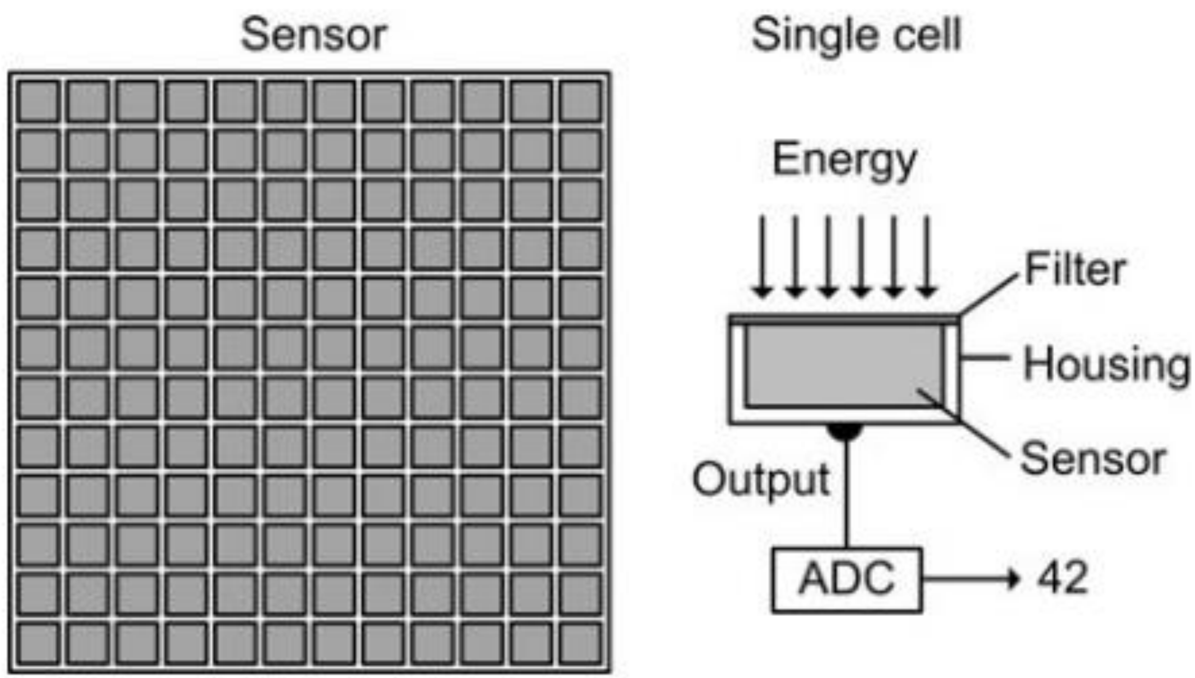
An **aperture** (or diaphragm): a small circular hole in the case that lets in light for a short period.

A **shutter** mechanism: a spring-loaded set of overlapping blades, that open to let light in through the aperture for a precise amount of time before closing up again. (James Bond opening)

Lenses in front of the shutter: **scale down** the large, incoming image of the world so it fits into a much smaller area of film; **concentrate the incoming light energy** so the image forms on the film more quickly and the camera can be used in darker conditions**; bring the light rays into a sharp focus**; **minimize the distance between the aperture and the film** so cameras can be small and portable;

A roll or piece of **film** (on the back wall of the camera directly opposite the shutter)

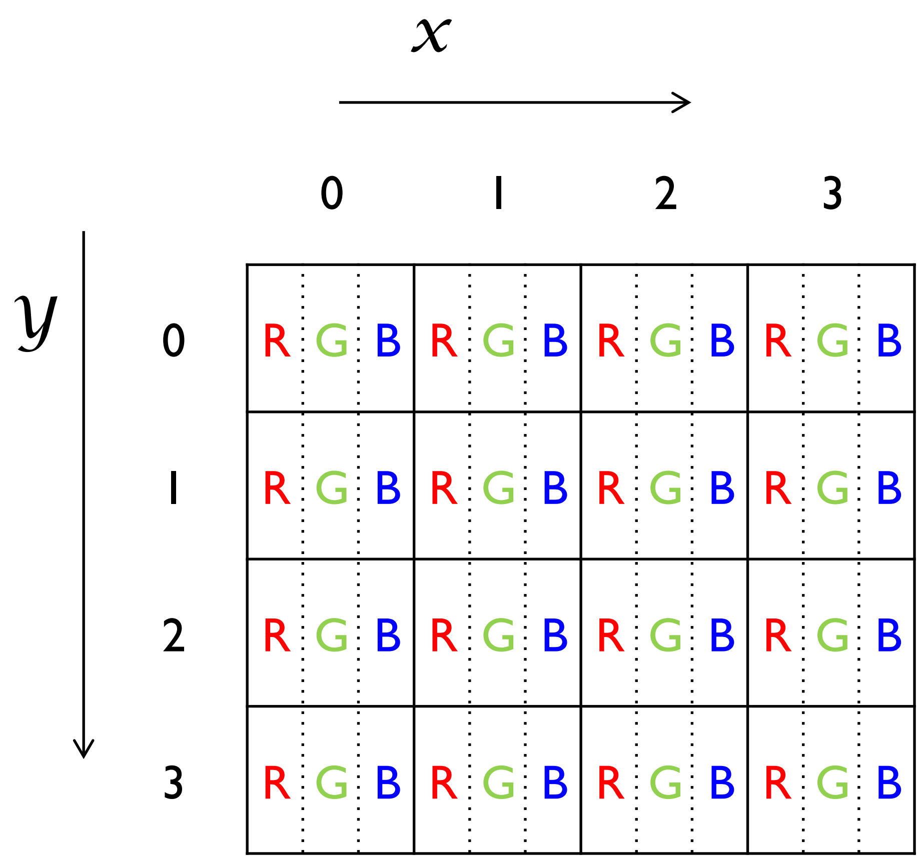
Digital Image -Pixels



Digital Image - Color

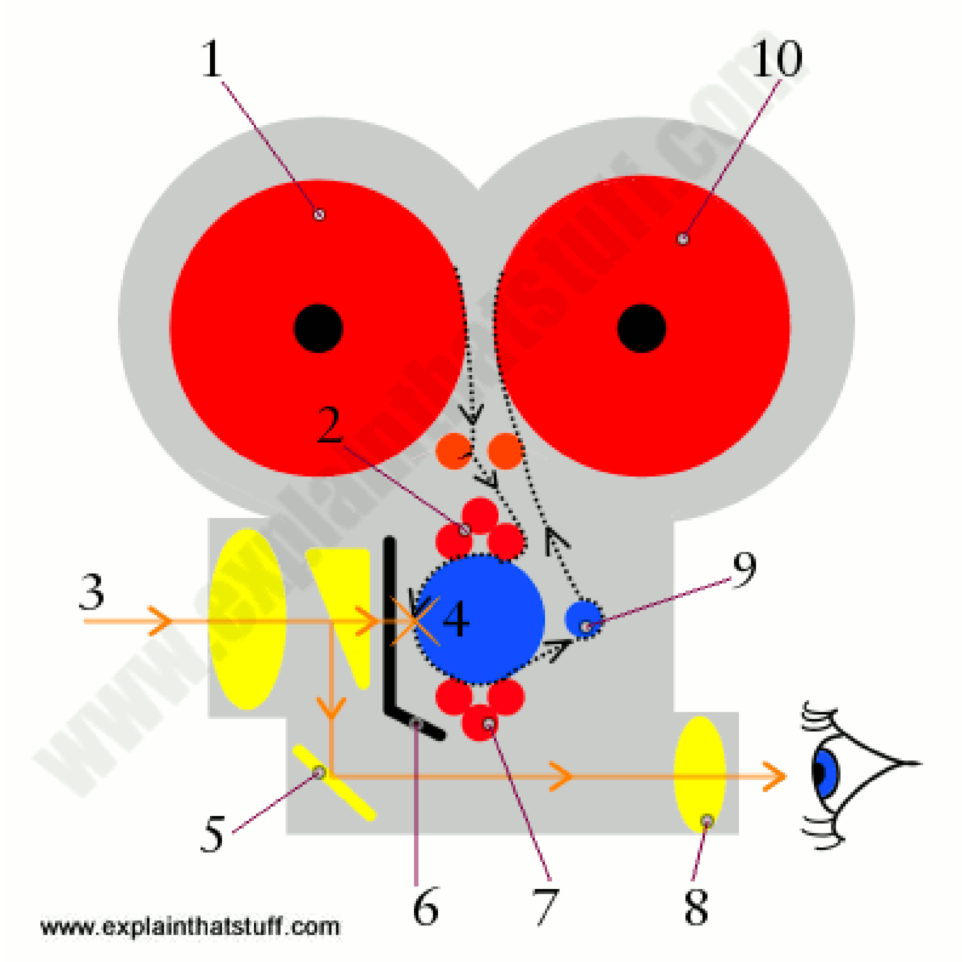
Red, Green, and Blue

RGB Color: 8 bit x 3
= 24 bit



Moving Images – Input

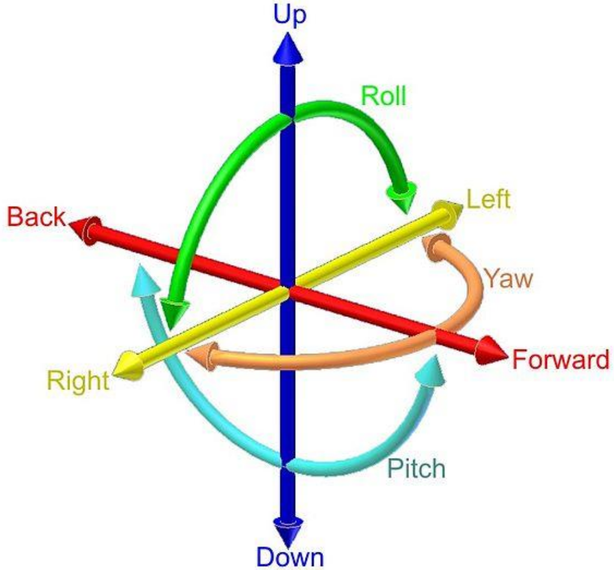
-Film Cameras (like photo cameras but film is moving, shutter opens 20 - 60 times per second)



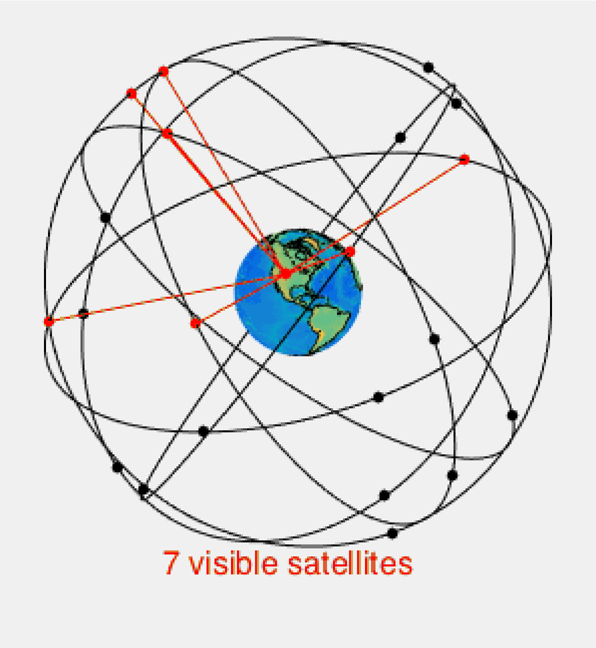
Inertia

Inertia is the resistance of any physical object to any change in its state of motion; this includes changes to its speed, direction or state of rest.

Degree of Freedom (DoF): independent dimension of measurement.
3DoF orientation + 3DoF position = 6DoF



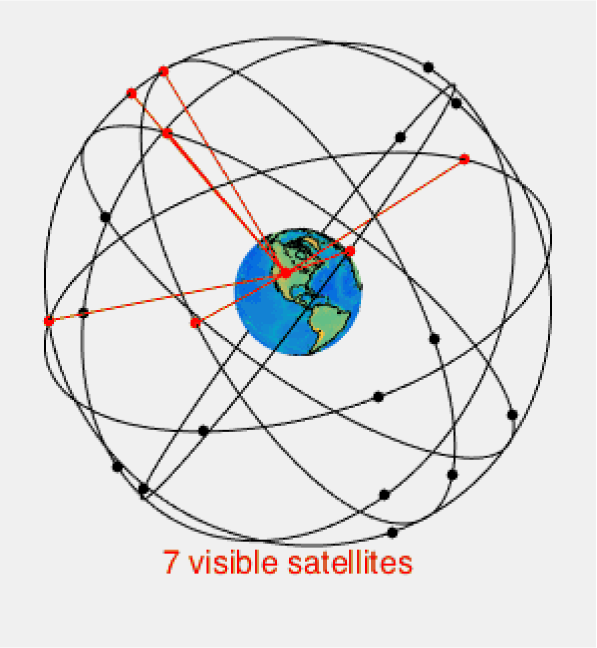
GPS – Global Position System



https://en.wikipedia.org/wiki/Global_Positioning_System

The GPS module receives timestamped signals from orbiting satellites. By calculating the time difference between signal transmission and reception from multiple satellites, the phone calculates its exact coordinates (trilateration).

GPS – Global Position System



https://en.wikipedia.org/wiki/Global_Positioning_System

https://www.youtube.com/watch?v=4O3ZVHVFhes

https://www.youtube.com/watch?v=FU_pY2sTwTA

Magnetometer (Compass)

The magnetometer measures the strength and direction of magnetic fields, including the Earth's magnetic field. This allows the phone to determine its orientation relative to North, South, East, and West.

https://www.youtube.com/watch?v=Ijfa1R9JXWk

Gyroscope

The gyroscope measures angular velocity—how fast the phone is rotating around an axis. While the accelerometer measures linear motion, the gyroscope measures rotation.

https://www.youtube.com/watch?v=JnKIoSdUJLo

Accelerometer

An accelerometer measures acceleration forces in three dimensions (x, y, and z axes). It primarily detects changes in velocity (speeding up or slowing down).

https://youtu.be/KZVgKu6v808?t=31s

Ambient Light Sensor

The ambient light sensor acts as the "eyes" of the smartphone regarding brightness. Similar to how the human pupil constricts in bright light, this sensor detects the intensity of the surrounding light and automatically adjusts the screen's brightness.

Proximity Sensor

Usually located near the earpiece (or under the display in newer flagship phones), this sensor consists of an infrared LED and an infrared light detector.

When you hold the phone up to your ear during a call, the sensor detects the reflection of the invisible infrared light off your skin. It then instantly turns off the display.

MEMS and IMU

MEMS - Microelectromechanical systems

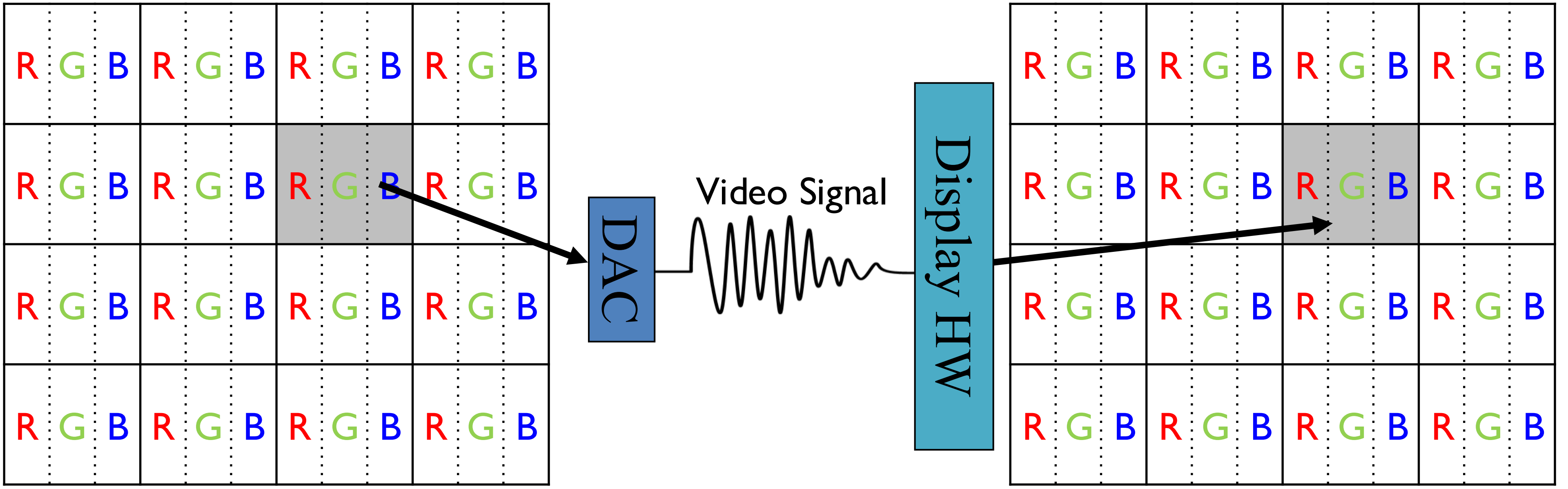
https://youtu.be/eqZgxR6eRjo?t=25s

IMU – Inertial Measurement Unit: a combination of (3) accelerometers and (3) gyroscopes, sometimes also (3) magnetometers.

Mobile Devices – Output

Frame Buffer

Values in memory to be displayed



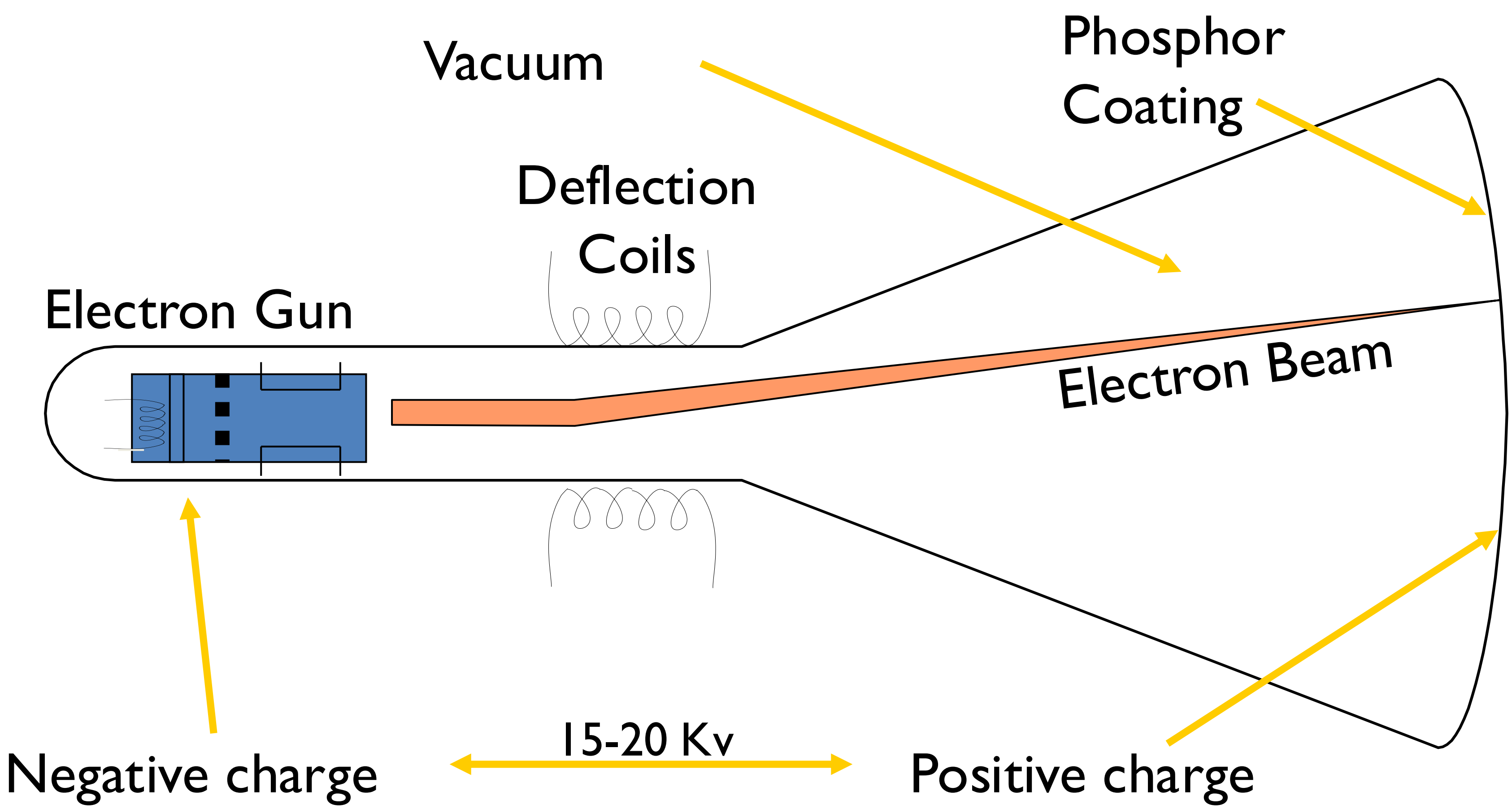
Visual Output – Display Devices

CRT (Cathode Ray Tube) Monitors

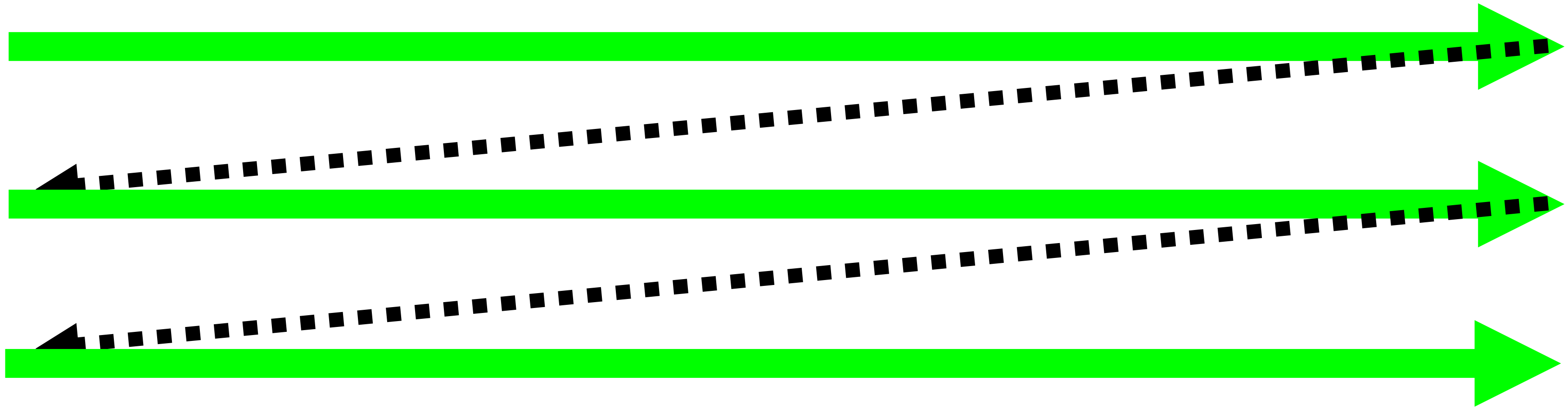
Hung on as dominant technology
for a long time

- Because CRT TVs were a mass market consumer item
- Many many years of work to make them cheap
- Displaced only when TVs started using LCDs

How a CRT works



Creating Pixels on Screen



“Raster” lines across screen

Modulate intensity along line (in spots) to get pixels

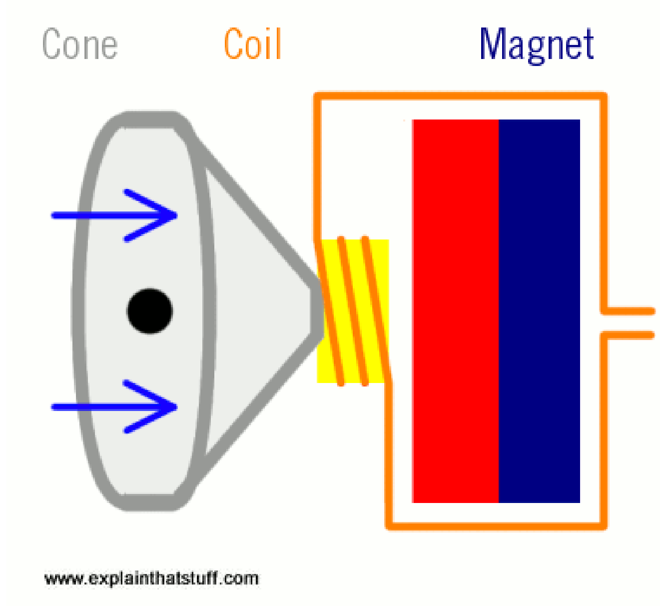
- Color requires 3 electron guns and 3 spots of phosphor for each pixel (quite complex)

http://www.bit-tech.net/hardware/2006/03/20/how_crt_and_lcd_monitors_work/2.html

Audio - Output

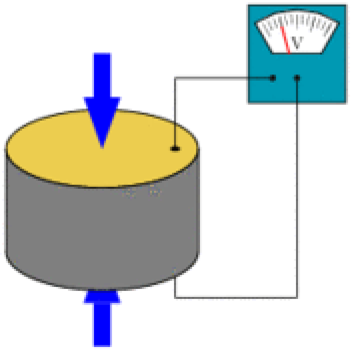
Loudspeackers: “reserved microphone”

Digital: use DAC to transform digital into electrical current



Vibration

Piezoelectric actuators



https://en.wikipedia.org/wiki/Piezoelectricity

Eccentric Rotating Mass vibration motors

http://www.sensorwiki.org/doku.php/actuators/eccentric_rotating_mass_erm_motor